

Declan Chan

Basics

Name: Declan Chan

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Education

BASc	University of Toronto , Engineering Science	2025 – present
	Dean's Honours List (all semesters)	
	Maple High School , IB Diploma Programme	2021 – 2025
	- IB Score: 45/45 - OSSD Top 6: 99.67% - Director's Achievement Award	

Technical Skills

Languages: Python, C, MATLAB, Bash

Design: SolidWorks, OnShape

Tools: Git, Linux, Slurm

Experience

Autonomous Discovery of Alloys , Machine Learning Researcher	2026 – present
- Designed experiments to evaluate the robustness of machine learning force fields. - Benchmarked MLFF predictions against density functional theory calculations. - Performed large-scale experiments on Canada's Fir HPC cluster.	1 year
University of Toronto Aerospace Team , Space Systems Engineer	2026 – present
- Developed software for CubeSat attitude determination and control systems. - Developed Python modules for attitude dynamics and flight simulation. - Simulated magnetic environments using a Helmholtz cage and MATLAB.	1 year
Robotics for Space Exploration , Mechanical Design Engineer	2025 – 2026
- Designed robotic arm components using CAD for 3D modelling. - Performed finite element analysis in SolidWorks Simulation to evaluate designs.	1 year
Northern Karate Schools , Senior Instructor	2021 – present
- Yondan: Certified 4th Degree Black Belt. - Gold medal at the Toronto International Tournament of Martial Arts Champions. - Teach 100+ students each week through group and private lessons.	5 years

Projects

Attacks Against Machine Learning Force Fields	2026
- Developed open-source software to generate first-order adversarial attacks against MLFFs. - Built Python/PyTorch pipelines to process Materials Project and LiCOHPF datasets. - Automated ASE relaxation, perturbation, and contour exploration workflows using Bash and Slurm.	

GenAI Genesis Hackathon	2026
• Developed an interactive platform visualizing human disease progression. • Implemented a RAG system trained on PubMed data via IBM watsonx.ai. • Built an AI-powered 3D web application with React and three.js.	
CIV102 Bridge Project	2025
• Optimized structural performance and prototyped a matboard bridge using Python. • Top 1 in the cohort and 2nd in the class of 2025.	
Astrophysics Research Project	2024 – 2025
• Conducted data analysis and literature review on Yarkovsky effect orbital perturbations. • Modelled thermal-radiation orbital drift and trajectories of several near-Earth asteroids.	
NFT Generation	2021 – 2022
• Used the HashLips Art Engine to generate layered NFT artwork from image assets. • Created and deployed NFTs on OpenSea.io using the Polygon Ethereum blockchain.	
ECs	
Rubik's Cube Competitions	2022 – 2025
• Top 90 in Canada for 3x3x3 fewest moves. • Volunteered as a judge for the World Cube Association. • Club President who taught beginners and organized school-wide competitions.	
Varsity Volleyball	2022 – 2025
• Team Captain in Grade 10 and 12. • Tier 1 YRAA Championship junior varsity 2nd place. • Tier 1 YRAA Championship senior varsity 2nd place.	
Rep Volleyball	2019 – 2023
• Team Captain and 2X MVP. • Silver medal in Canada's 2023 Youth Nationals.	
Music Performance	2014 – 2023
• Theory and Instrumental: Level 10 RCM comprehensive certificate (First Class Honors). • Level 10 RCM practical piano certificate (First Class Honors with Distinction). • Level 8 RCM practical violin certificate (First Class Honors with Distinction).	